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Artifactual hyperpolarization during extracellular electrical stimulation: Proposed mechanism of high-rate neuromodulation disproved

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ABSTRACT

Background: Kilohertz-frequency electric field stimulation (kEFS) applied to the spinal cord can reduce chronic pain without causing the buzzing sensation (paresthesia) associated with activation of dorsal column fibers. This suggests that high-rate spinal cord stimulation (SCS) has a mode of action distinct from conventional, parasthesia-based SCS. A recent study reported that kEFS hyperpolarizes spinal neurons, yet this potentially transformative mode of action contradicts previous evidence that kEFS induces depolarization and was based on patch clamp recordings whose accuracy in the presence of kEFS has not been verified.

Objectives: We sought to elucidate the basis for kEFS-induced hyperpolarization and to validate the effects of kEFS observed in patch clamp recordings by comparing with independent optical methods.

Methods: Using patch clamp electrophysiology and voltage-sensitive dye (VSD) imaging, we measured the response to kEFS applied *in vitro* to hippocampal and spinal neurons.

Results: The kEFS-induced hyperpolarization observed with current clamp recordings was corroborated by VSD imaging and rheobase measurements in patched neurons. However, no hyperpolarization was observed when imaging unpatched neurons or when recording with a voltage-follower amplifier rather than with a patch clamp amplifier (PCA). We found that EFS induced an artifactual current in PCAs that was injected back into current clamped neurons. The artifactual current induced by single, charge-balanced EFS pulses caused modest hyperpolarization, but these unitary hyperpolarizations accumulated when EFS pulses were repeated at kilohertz frequencies.

Conclusion: Our results rule out hyperpolarization as the mechanism underlying kEFS-mediated analgesia and highlight the risk of recording artifacts caused by extracellular electrical stimulation.

Introduction

Extracellular electrical stimulation is widely used to activate neurons in experiments and for diverse clinical applications [1]. It is reasonably well understood how electric field stimulation (EFS) activates neurons [2–4]. Neurons can also be inhibited by EFS; specifically, EFS applied to peripheral nerves at frequencies between 10 and 100 kHz (and as low as 4 kHz in non-mammalian axons) has been shown to block spike propagation by causing depolarization and sodium channel inactivation [5–10]. Net inhibition can also occur indirectly if EFS-evoked spikes engage inhibitory neurons, as occurs in conventional low-frequency (40–60 Hz) spinal cord stimulation (SCS) where, consistent with the Gate Control Theory [11], spikes evoked in dorsal column fibers propagate antidromically into the spinal dorsal horn and engage inhibitory neurons that help block pain signals [12–14]. Recently, application of EFS at frequencies between 1 and 10 kHz (henceforth referred to as kEFS) to the spinal cord has emerged as a promising new form of SCS [15,16]. Unlike conventional SCS, which is associated with a buzzing sensation (paresthesia) caused by orthodromic

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propagation of the spikes evoked in dorsal column fibers, high-frequency SCS is paresthesia-free, which suggests that dorsal column fibers are not activated [17,18]. But nor is high-frequency SCS associated with the sensory deficits that would occur if spike propagation is blocked. The mechanism of action of high-frequency SCS thus remains unclear and has become the focus of intense study given the clinical efficacy of this treatment [19,20].

Recent experiments suggested that kEFS hyperpolarizes spinal neurons [21]. Those results, if true, demonstrate a novel form of *direction inhibition* that could be harnessed to treat a multitude of neurological disorders and would represent a major breakthrough in elucidating how high-rate SCS relieves pain. However, the experimental data reported by Lee et al. are inconsistent with simulations [22] and with studies on nerve block (see above) that predict, if anything, that depolarization should occur. This discrepancy suggests that either the simulations or the experiments are flawed. With respect to potential experimental flaws, patch clamp amplifiers have been reported to exhibit recording artifacts when used in current clamp mode [23,24]. The effects on voltage measurement are usually quite minor, but, to the best of our knowledge, no one has tested whether EFS exacerbates this artifact in PCAs or causes some other aberration. Ruling out such artifacts and reconciling the experimental data with simulations is critical before moving forward.

We thus sought to verify the original report of kEFS-induced hyperpolarization [21] and elucidate how that hyperpolarization occurs using electrophysiological *and* optical methods. After reproducing the hyperpolarization, inconsistencies revealed by optical recording prompted deeper investigation that ultimately revealed the hyperpolarization to be caused by an artifactual current originating from the PCA during EFS. Refutation of kEFS-induced hyperpolarization as a novel form of direct inhibition is disappointing for neuromodulation applications, but awareness of the artifact we have uncovered is critical for anyone studying the cellular effects of extracellularly applied electric fields.

Materials and methods

Cell culture and slice preparation

All experimental procedures were approved by the Animal Care Committee at The Hospital for Sick Children. To culture hippocampal neurons, a pregnant Sprague-Dawley rat (Charles River, Montreal) was anesthetized with 4% isoflurane and killed by cervical dislocation. Hippocampi from 8 to 12 embryos (17–19 days gestation) were dissected, placed in sterile Hank's solution at 4 °C, and mechanically dissociated by trituration. Hippocampal neurons were plated onto poly-D lysine-coated coverslips with Neurobasal™ media (Gibco 21103–049) supplemented with 1% fetal bovine serum (FBS), B-27™ supplement (Thermo Fisher 17504–044) and 0.5 mM L-Glutamine (Gibco 25030–081) at a culture density of $<1 \times 10^6$ cells/mL. Coverslips were maintained for up to 3 weeks in a 37 °C incubator at 7% CO_2 with maintenance media (plating media without FBS) changed every 3–4 days. Neurons were used for experiments at 14–21 days *in vitro* (DIV). Slices were prepared as described previously [25,26].

Electrophysiology

The recording chamber was perfused at ~2 ml/min with oxygenated (95% O_2 and 5% CO_2) artificial cerebrospinal fluid (ACSF). To block fast synaptic transmission, ACSF was supplemented with (in μM) 10 bicuculline methiodide, 10 CNQX and 40 D-AP-5 (HelloBio, Princeton, NJ). We patched neurons visualized on coverslips (cultures) or in lamina I/II of spinal cord slices or in the CA1 region of hippocampal slices. Neurons were recorded in the whole cell configuration with >70% series resistance compensation using an Axopatch 200B amplifier or an Axoclamp 2B amplifier (Molecular Devices, Sunnyvale, CA). Patch pipettes were pulled from borosilicate glass capillaries (WPI) and were filled with an intracellular solution composed of (in mM): 125 $KMeSO_4$, 5 KCl, 10 HEPES, 2 $MgCl_2$, 4 ATP, 0.4 GTP; pH was adjusted to 7.2 with KOH (pipette tip resistance 6–9 mΩ). All recordings were at room temperature (~22 °C). For most experiments, traces were low pass filtered at 100 kHz and digitized at 500 kHz using a CED 1401 computer interface (Cambridge Electronic Design, Cambridge, UK). To standardize across neurons, pre-stimulus membrane potential was adjusted to −65 mV (after correction for a liquid junction potential of −9 mV) using tonic current injection.

Voltage-sensitive dye (VSD) imaging

Cultured neurons 14–21 DIV were stained using a FluoVolt Membrane Potential Kit™ (Life Technologies, Carlsbad, CA) [27]. FluoVolt dye was diluted in the PowerLoad concentrate and Live Cell Imaging Solution (LCIS) (ThermoFisher Scientific), supplemented with 10 mM glucose to a final concentration 75% of that recommended by the manufacturer. Coverslips were washed 3x in room temperature LCIS before undergoing a 15 min incubation in the diluted FluoVolt solution followed by 3x glucose supplemented LCIS washes. The VSD was excited with a 505 nm LED (Thorlabs, Newton, NJ) at an intensity of 0.68 mW/mm^2 and was imaged with a 40x water immersion lens (0.75 NA) and Zeiss filter set 46HE (excitation, 500/25; emission, 535/30). Neurons were imaged during 500 ms-long light exposures (separated by 700 ms-long dark periods) using a NeuroCCD-SM256 camera at an acquisition rate of 50 Hz on maximal gain (30x) (RedShirt Imaging, Decatur, GA). Image analysis was performed with Neuroplex software (RedShirt Imaging). Calibration curves were generated by measuring fluorescence intensity changes (ΔF) during current injection steps (in the absence of kEFS) while recording membrane voltage in current clamp thus allowing ΔF measurements (in arbitrary units reflecting pixel intensities) to be converted directly to ΔV on a cell-by-cell basis.

Electric field stimulation

Extracellular electrical stimulation was delivered through a pair of preclinical leads (Boston Scientific, Valencia, CA) placed 10 mm apart in the recording chamber (Fig. 1A). One lead served as the anode and the other as the cathode. Lead impedance was tested before each experiment and was consistent across the study. Constant current was distributed equally to the four platinum contacts on each lead using a therapy-grade Precision Spectra™ External Trial Stimulator (ETS) from Boston Scientific [28] that had undergone a non-commercial firmware change to enable the delivery of waveforms with frequencies up to 10 kHz. The ETS is battery powered and remote controlled by the Precision Spectra™ Clinician Programmer tablet and meets IEC60601 standards for electrical isolation, which means that, except for the stimulator leads in the recording chamber, the entire stimulation system is electrically isolated from the recording equipment. The ETS was further shielded with a grounded tinfoil sheet; grounding the sheet to the isolation table (to which other equipment was grounded) or to a separate ground made no difference. Indeed, ETS shielding served only to reduce transients near the onset and offset of EFS that originated from commands sent via remote control from the programmer. Simulations predicted (Sup. Fig. S1) and measurements of bath voltage confirmed (Fig. 1B) that equal distribution of constant current to the four contacts on each of the two, parallel

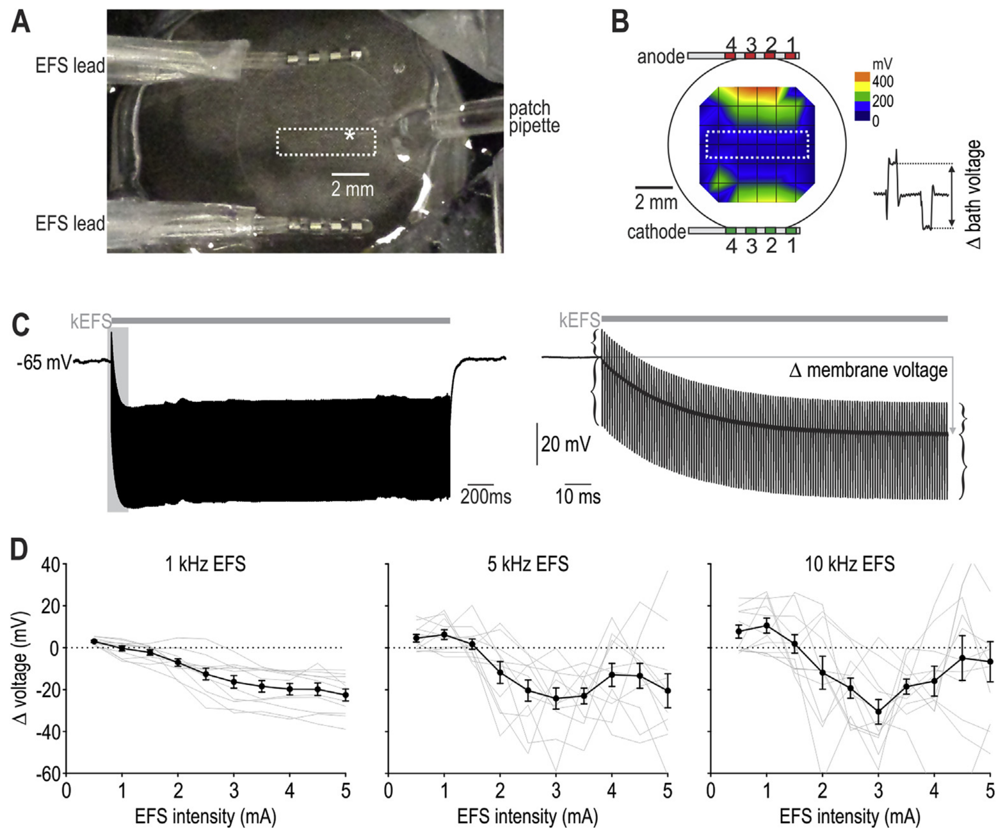

Fig. 1. Patched neurons are hyperpolarized by kEFS. **(A)** Photograph showing arrangement of two 4-contact leads to deliver EFS. **(B)** Sample measurement of electric field for 5 mA EFS based on peak-to-peak voltage deflections measured in the bath at multiple points along a grid. Neurons were recorded in the central region (dotted rectangle). **(C)** Sample current clamp recording showing typical voltage response to 5 mA, 1 kHz EFS. Right panel shows enlarged view of shaded region on left. Membrane potential during kEFS was measured during interval between artifacts and is not always vertically centered between the peaks and troughs of artifacts (see brackets). **(D)** Summary data from 11 neurons tested with all combinations of EFS intensities and frequencies. Black curve shows mean ± sem. Gray curves show data from individual neurons.

oriented leads produced a uniform electric field. A biphasic charge-balanced waveform with 40 μs-long pulses separated by a 10 or 100 μs-long interpulse interval (for EFS < or >1.2 kHz, respectively) was applied at 1, 5 or 10 kHz. The waveform shape and charge balance were confirmed with recordings of the bath voltage (see inset on Fig. 1B). To measure the unitary hyperpolarization, we studied the first response to the same waveform (40 μs pulses with 100 μs interpulse intervals) applied at 150 Hz.

Statistical analysis

Statistical analysis was conducted using Sigmaplot v11. Distributions were tested for normality using the Kolmogorov−Smirnov test. When neurons were re-tested under different conditions (e.g. different EFS frequencies or amplitudes), a paired *t*-test or repeated measures ANOVA was applied. For comparing between groups (e.g. between amplifiers or pipette shielding conditions), an unpaired *t*-test was applied. Convolutions were carried out in Matlab.

Results

EFS-induced hyperpolarization is observed in current clamp recordings

Current clamp recordings in cultured hippocampal neurons revealed robust hyperpolarization during kEFS (Fig. 1C); spinal neurons behave the same way (see below). Data from 11 neurons tested with all EFS parameter combinations − frequencies of 1, 5, and 10 kHz with amplitudes from 0.5 to 5 mA − are summarized in Fig. 1D. Group averages (±SEM) are shown in black and data from individual cells are shown in gray. Stimulus amplitude had a significant effect ($F_{9,10} = 8.724$, $p < .001$) whereas frequency did not ($F_{2,10} = 0.897$, $p = .424$), although there was a significant interaction between amplitude and frequency ($F_{18,10} = 1.795$; $p = .029$, two-way repeated measures ANOVA). Since 1 kHz EFS evoked the most consistent responses, most subsequent testing focussed on this frequency.

To demonstrate the inhibitory effect of this hyperpolarization, we tested whether kEFS-induced hyperpolarization increased rheobase, the minimum injected current required to evoke spiking. Rheobase was measured before and during kEFS by injecting a series of current steps with increasing amplitudes into the neuron (Fig. 2A). Rheobase was significantly increased by 5 mA, 1 kHz kEFS (Fig. 2B, $t = 9.114$, $p < .001$, paired *t*-test), thus confirming that spiking is inhibited by kEFS-induced hyperpolarization. This confirmed that kEFS-induced hyperpolarization is not illusory, but the possibility remained that the act of measuring voltage might introduce the hyperpolarization (i.e. an observer effect); in other words, the voltage change is measured correctly but might arise because of how it is measured (see below).

kEFS-induced hyperpolarization occurs only in patched neurons

To quantify kEFS-induced hyperpolarization using a method independent of the PCA, neurons were loaded with a voltage-sensitive dye (VSD) so that voltage could be measured optically (Fig. 3A and B). VSD-based measurements confirmed that 5 mA, 1 kHz EFS caused significant hyperpolarization in patch clamped

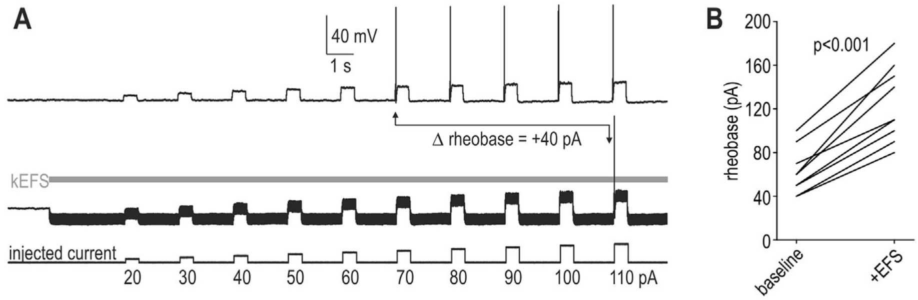

Fig. 2. kEFS-induced hyperpolarization inhibits spiking. (A) Sample traces show determination of rheobase at baseline (top) and during kEFS (bottom) based on the weakest current injection step able to evoke spiking. **(B)** Summary of rheobase at baseline and during 5 mA, 1 kHz EFS for 9 neurons. Rheobase was significantly increased by kEFS ($t = 9.114$, $p < .001$, paired t-test).

neurons (Fig. 3C left panel; $t = 7.345$, $p < .001$, one-sample t-test). Electrophysiological and VSD-based voltage measurements were tightly correlated ($r^2 = 0.82$) and the slope of the regression line (0.90 ± 0.17) did not differ significantly from the expected slope of 1 ($t = 0.59$, $p = .58$, one-sample t-test) (Fig. 3C right panel).

Unlike the kEFS-induced change in fluorescence illustrated in Fig. 3A, VSD imaging conducted in the same neuron, but before patching (with the patch pipette positioned in the bath, near but not on the neuron), revealed no kEFS-induced fluorescence change (Fig. 4A). Comparing VSD-measured hyperpolarization induced by 5 mA, 1 kHz EFS applied before patching, during current clamp recording, and again after removal of the patch pipette revealed hyperpolarization only when neurons were patched (Fig. 4B, left panel). This led us to hypothesize that the patch pipette acts as an antenna that converts the varying electric field into current that flows into the neuron. From this, we predicted that shielding the patch pipette with Sylgard would attenuate kEFS-induced hyperpolarization. As predicted, there was a significant interaction between shielding status and patching status on VSD-measured hyperpolarization ($F_{2,13} = 10.433$, $p < .001$; two-way repeated measures ANOVA), with significantly greater kEFS-induced hyperpolarization occurring with unshielded pipettes (-21.8 ± 1.4 mV; mean $\pm$ SEM) than with shielded pipettes (-10.0 ± 1.5 mV) ($p < .001$, Student-Newman-Keuls test) (Fig. 4B). Electrophysiological measurements using a PCA corroborated these findings (Fig. 4C, $t = 3.622$, $p = .003$, t-test).

To ensure that the artifactual nature of kEFS-induced hyperpolarization was not limited to 1 kHz EFS, VSD measurements were conducted in additional cells tested with 3.5 mA, 10 kHz EFS. As with 1 kHz EFS, patched neurons exhibited kEFS-induced voltage changes (Fig. 4D). The effect varied across neurons in a manner consistent with variability reported in Fig. 1D, but VSD and electrophysiological measurements were tightly correlated within each neuron (Fig. 4E). VSD imaging in an additional ten unpatched neurons showed no significant fluorescence changes induced by 10 kHz EFS applied at any intensity (Fig. 4F, $p > .05$ for all one-sample t-tests even before correction for multiple comparisons). Fluorescence changes were not converted to voltage changes because calibration curves require patching, but based on typical calibration curves, deviations in Fig. 4F correspond to <1 mV. The

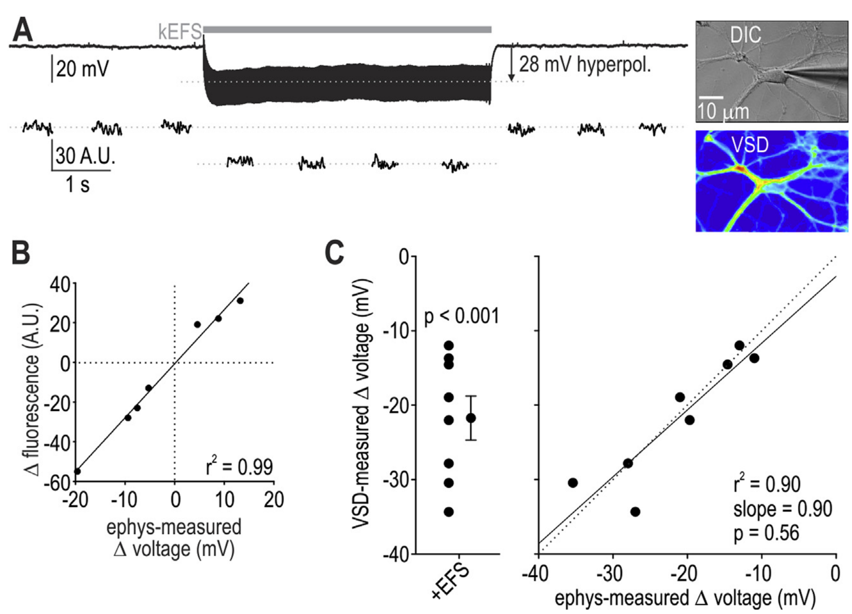

Fig. 3. VSD imaging confirms kEFS-induced hyperpolarization of patched neurons. (A) Sample electrophysiology trace (top) and simultaneous VSD measurements (in arbitrary units, AU) from 500 ms-long epochs before, during, and after EFS (bottom). Imaging was conducted in epochs to help minimize bleaching and phototoxicity. The kEFS-induced fluorescence change was quantified as the difference between fluorescence averaged across epochs during kEFS vs fluorescence averaged across the two epochs before and two epochs after kEFS. Insets show differential interference contrast (DIC) image (top) and pseudo-colored VSD image (bottom). **(B)** Sample calibration curve based on fluorescence measurements during simultaneous current clamp recordings, with voltage changes induced by current injection steps. **(C)** Summary of VSD-measured hyperpolarization during 5 mA, 1 kHz EFS in 8 neurons (left); hyperpolarization was significant ($t = 7.345$, $p < .001$, one-sample t-test). Right panel shows correlation between VSD- and electrophysiology-measured voltage changes during kEFS.

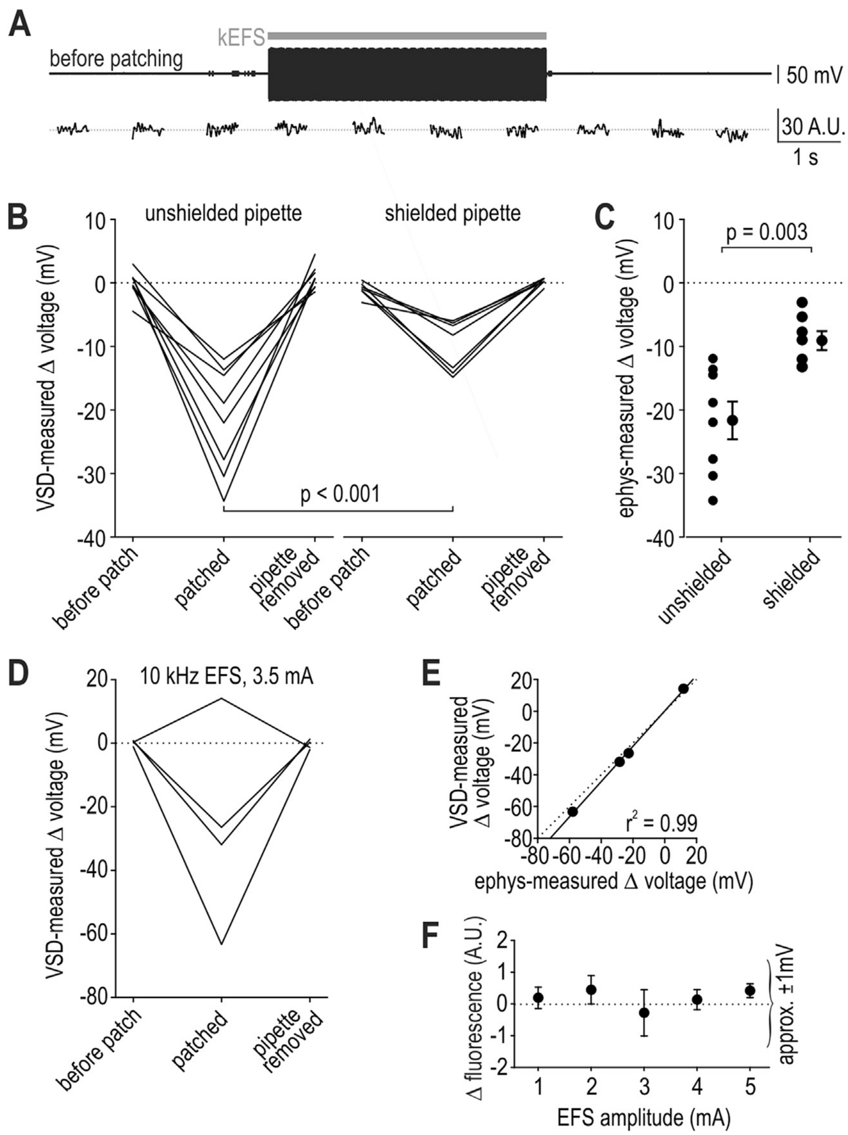

Fig. 4. VSD imaging reveals no kEFS-induced hyperpolarization in unpatched neurons. **(A)** Sample VSD traces from same neuron as in Fig. 3A but before patching. Top trace shows EFS-induced fluctuations in bath voltage measured with the patch pipette positioned just above the neuron. **(B)** Summary of VSD-measured voltage changes induced by 5 mA, 1 kHz EFS tested before, during, and after patch clamp recordings; the patch pipette was removed from the neuron after electrophysiology testing to conduct the final VSD measurement. Testing was performed on 7 neurons recorded with pipettes shielded with Sylgard (right) for comparison with measurements from 8 neurons recorded with unshielded pipettes (left). kEFS-induced hyperpolarization was evident only when neurons were patched, and was significantly less with shielded pipettes ($p < .001$; Student-Newman-Keuls test). **(C)** Summary of electrophysiology-measured hyperpolarization in same neurons as in B. Here again, kEFS-induced hyperpolarization was significantly less with shielded pipettes ($t = 3.622$, $p = .003$, t-test). **(D)** Additional neurons tested with 3.5 mA, 10 kHz EFS exhibited kEFS-induced VSD fluorescence changes only when patched. **(E)** VSD-measured voltage changes were tightly correlated with electrophysiology-measured voltage changes. Solid line shows linear regression ($r^2 = 0.99$) for comparison with equivalence between the two types of measurements (dotted line). **(F)** VSD measurements (mean fluorescence change ± sem) in 10 unpatched neurons tested with 10 kHz EFS between 1 and 5 mA. Fluorescence intensity was not significantly altered from baseline at any intensity ($p > .05$, one-sample t-tests even before correction for multiple comparisons).

lack of kEFS-induced hyperpolarization in unpatched neurons is also consistent with computer simulations (Sup. Fig. S2).

kEFS-induced hyperpolarization depends on the type of amplifier and recording mode

Reports that patch-clamp amplifiers (PCAs) can cause artifacts during current clamp recordings [23,24] led us to suspect that our Axopatch 200B PCA was implicated in converting the electric field into a hyperpolarizing current. We therefore repeated electrophysiology-based measurements of EFS-induced hyperpolarization using an Axoclamp 2B, which operates as a true voltage-follower amplifier (VFA). Fig. 5A shows the response of a typical neuron recorded with this VFA during 5 mA, 1 kHz EFS. Measurements conducted with the VFA revealed no kEFS-induced hyperpolarization (Fig. 5B; 0.1 ± 0.1 mV, $t = 0.574$, $p = .584$, one sample t-test) and no change in rheobase (Fig. 5C; $t = 0.0$, $p = 1.0$, paired t-test). To verify that kEFS-induced hyperpolarization was not unique to the particular PCA we had used, we measured the effects of 5 mA, 1 kHz EFS in 9 additional neurons using a different Axopatch 200B that was otherwise connected and used in the same way. We observed −22.1 ± 4.3 mV hyperpolarization, which was not significantly different from the average hyperpolarization of −22.2 ± 2.0 mV measured using the original Axopatch 200B

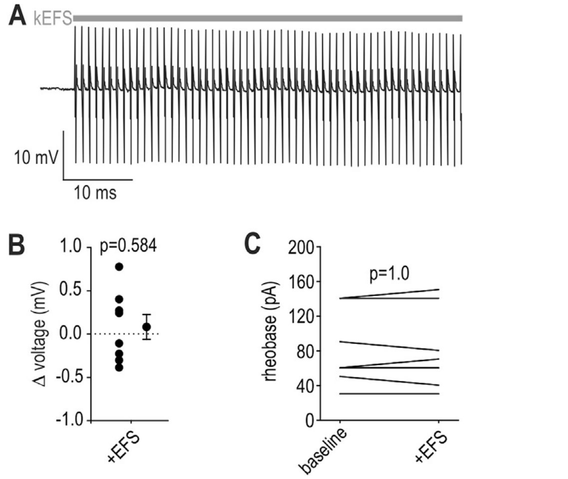

Fig. 5. Neurons recorded with voltage-follower amplifier (VFA) are not hyperpolarized by kEFS. (A) Sample recording with a VFA during 5 mA, 1 kHz EFS. **(B)** Summary of voltage change induced by 5 mA, 1 kHz EFS in 8 neurons. Average change of 0.1 ± 0.1 mV was not significantly different from 0 ($t = 0.574$, $p = .584$, one sample t-test). **(C)** Nor did these neurons exhibit any kEFS-induced change in rheobase ($t = 0.0$, $p = 1.0$, paired t-test).

($n = 19$ after pooling across experiments, $t = 0.0302$, $p = .976$, t-test). Further testing revealed that *Iclamp fast* and *Iclamp normal* settings on the Axopatch 200B yielded equivalent kEFS-induced hyperpolarization (data not shown). These data suggest that the artifact is common to PCAs but is not present in VFAs, especially since the two types of amplifiers were connected and used in the same way.

Finally, to assess if the recording mode and associated circuitry were critical, we used a PCA to conduct back-to-back voltage clamp and current clamp recordings in three additional neurons during 2 mA, 1 kHz EFS; weaker EFS was used to avoid saturating the voltage clamp signal. Using each neuron's input resistance to calculate the current responsible for hyperpolarization, current clamp recordings revealed hyperpolarization consistent with a 44.7 ± 6.4 pA outward current, which is significantly different from 0 ($t = 12.0$, $p = .007$, one-sample t-test). Voltage clamp recordings revealed a 14.4 ± 8.5 pA inward current, which is not significantly different from 0 ($t = 1.7$, $p = .23$). The discrepancy between the two measurements is significant ($t = 5.339$, $p = .033$, paired t-test), demonstrating that the artifact is limited to the current clamp mode of PCAs.

This last observation, namely that recordings conducted with the same Axopatch 200B amplifier (PCA) did or did not exhibit the artifact depending on the recording mode (current clamp vs. voltage clamp, respectively), argues against there being any defect in the stimulator (e.g. DC leak). The proper functioning of the stimulator is further supported by the lack of hyperpolarization in unpatched neurons (see Fig. 4) or in neurons recorded with an Axoclamp 2B amplified (VFA) (see Fig. 5). Instead, all of the data point to an artifactual hyperpolarization introduced by the PCA when operating in current clamp mode.

kEFS-induced hyperpolarization of neurons in acute slices

Like in cultured neurons, pyramidal neurons tested in acute hippocampal slices exhibited kEFS-induced hyperpolarization (Fig. 6A) and the associated change in rheobase (Fig. 6B) when recorded with a PCA but not when recorded with a VFA. Likewise, amongst dorsal horn neurons recorded from lamina I and II of

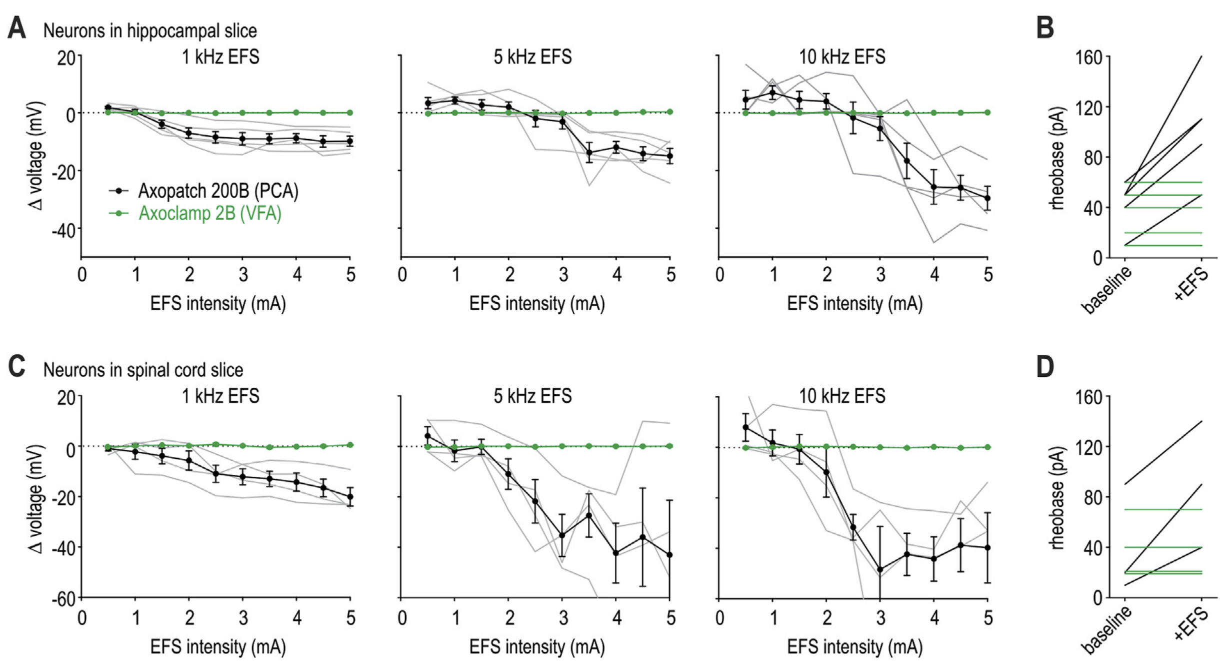

Fig. 6. Neurons in slice are hyperpolarized by EFS only if recorded with a patch clamp amplifier (PCA). (A) Comparison of kEFS-induced hyperpolarization in pyramidal neurons recorded from hippocampal slices with a PCA (black, $n = 5$) or a VFA (green, $n = 5$). Gray curves show data from individual neurons recorded with PCA. Hyperpolarization was observed only when recording with a PCA. **(B)** Rheobase was significantly increased by 5 mA, 1 kHz EFS in neurons recorded with a PCA (black, $t = 5.00$, $p = .008$, paired t-test) but not in neurons recorded with a VFA (green, $t = 0$, $p = 1.0$, paired t-test). **(C)** Similarly, lamina I and II neurons recorded from spinal cord slices with a PCA (black, $n = 4$) exhibited kEFS-induced hyperpolarization whereas no voltage change was seen when using a VFA (green, $n = 4$). **(D)** Rheobase was significantly increased by 5 mA, 1 kHz EFS in neurons recorded with a PCA (black, $t = 4.33$, $p = .049$, paired t-test) but not in neurons recorded with a VFA (green, $t = 0$, $p = 1.0$, paired t-test). In one neuron recorded with a PCA, the recording was lost before rheobase testing was completed.

spinal cord slices, kEFS-induced hyperpolarization (Fig. 6C) and the associated change in rheobase (Fig. 6D) were observed with the PCA, but not with the VFA. These results confirm that kEFS-induced hyperpolarization is not limited to a particular cell type or experimental preparation but, rather, is linked to the electrophysiological equipment.

Data presented above demonstrate that kEFS-induced hyperpolarization is an artifact rather than a novel basis for direct inhibition. The true mechanism of action of kEFS remains to be unravelled. To that end, understanding this EFS-induced artifact is critical for reassessing past studies whose data may have been compromised by it and for designing future studies and instrumentation to avoid it. This is relevant not only for neuromodulation research, but for all fields in which extracellular electrical stimulation is used. Thus, the remainder of our study was dedicated to uncovering the basis for the PCA-associated artifact.

Mechanism of kEFS-induced hyperpolarization

Inspecting the voltage trajectory at the onset of kEFS reveals that each biphasic EFS pulse evokes a transient hyperpolarization (arrows in bottom panel of Fig. 7A) and that these "unitary" hyperpolarizations appear to summate when EFS pulses are repeated at high rate (red curve in right panel). To characterize the unitary hyperpolarization, we measured hyperpolarization following the first EFS pulse (3 mA) delivered at 150 Hz — the lowest rate at which the stimulator could deliver therapy-grade biphasic rectangular pulses and found that the return of voltage to its baseline was well fit by a double exponential (Fig. 7B). For neurons tested with unshielded pipettes ($n = 6$), the fast time constant of the polarization decay was 215 ± 38 μs (mean ± SEM) with amplitude 3.09 ± 0.74 mV and the slow time constant was 9.19 ± 1.86 ms with amplitude of 1.52 ± 0.10 mV. For neurons tested with shielded pipettes ($n = 5$), the fast time constant was 293 ± 88 μs with amplitude 3.27 ± 0.48 mV and the slow time constant was 9.00 ± 2.22 ms with amplitude of 0.92 ± 0.26 mV. From here, we focussed on the slow component since the fast component was too brief to summate across the 1 ms interstimulus interval during 1 kHz EFS, and we found that the amplitude ($t = 2.318$, $p = .046$, t-test) but not the time constant ($t = 0.063$, $p = .951$, t-test) of the slow component was significantly reduced by pipette shielding.

To test whether the slow component of the unitary hyperpolarization summates to yield the cumulative hyperpolarization ascribed to kEFS, we convolved the unitary response with a 1 kHz event train. The voltage trajectory thus modeled (Fig. 7C) closely resembled experimental data (e.g. Fig. 7A). Furthermore, consistent with linear summation, the rate of cumulative hyperpolarization matched the decay time constant of the unitary hyperpolarization. Comparing cumulative hyperpolarization induced by 3 mA, 1 kHz EFS with hyperpolarization predicted from the summation of the unitary response revealed a strong linear correlation ($r^2 = 0.85$) whose slope of 0.77 ± 0.14 was not significantly different from the expected slope of 1 ($t = 1.66$, $p = .14$, one sample t-test) (Fig. 7D). This result demonstrates that the summation of unitary hyperpolarizations is sufficient to account for sustained hyperpolarization, which again argues against a contribution from other possible factors such as DC leak [29].

Next, we sought to explain how the unitary hyperpolarization arises from the PCA since kEFS-induced hyperpolarization occurs only in PCA-based current clamp recordings (see above). PCAs are designed around an *I-V* converter optimized for voltage clamp; current clamp is achieved by adding a positive feedback loop [30,31] (Fig. 8A left). Magistretti et al. showed that artifacts can arise in current clamp because of this circuitry [23,24], but they never tested the effects of extracellular EFS. Following their lead in monitoring output current (I_{out}), we observed that EFS induced nanoampere fluctuations in I_{out} (Fig. 8B bottom). The fast positive and negative components of I_{out} were equivalent, consistent with the symmetric EFS waveform and the symmetric fluctuation in bath voltage (V_b), again ruling out DC leak from the stimulator. A few

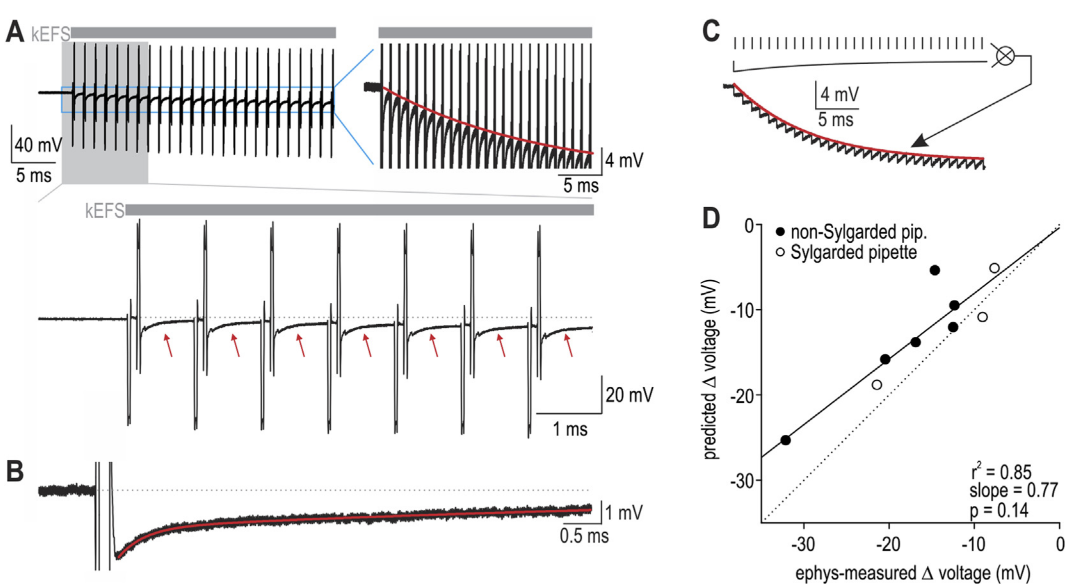

Fig. 7. Transient hyperpolarization after individual EFS pulses summate during kEFS. (A) Sample response recorded with PCA during 3 mA, 1 kHz EFS. Bottom panel shows horizontally expanded view of gray shaded region; red arrows highlight transient hyperpolarization after each EFS pulse. Right panel shows vertically expanded view of boxed region to illustrate summation of unitary hyperpolarizations (red curve). **(B)** Typical example of unitary hyperpolarization evoked by 3 mA EFS pulse. Voltage trajectory was well fit by a double exponential (red curve). **(C)** Convolving a 1 kHz pulse train with the slow component of the unitary hyperpolarization yielded cumulative hyperpolarization similar to that seen experimentally (compare with panel A). **(D)** Following the procedure in C, we used the unitary hyperpolarization measured from each neuron to predict cumulative hyperpolarization during 3 mA, 1 kHz EFS. Neurons with observed EFS-induced hyperpolarization <5 mV were excluded from analysis. The slope of 0.77 ± 0.14 did not differ significantly from the expected slope of 1 ($t = 1.66$, $p = .14$, one sample t-test), thus arguing that summation of the unitary hyperpolarization is sufficient to explain kEFS-induced hyperpolarization.

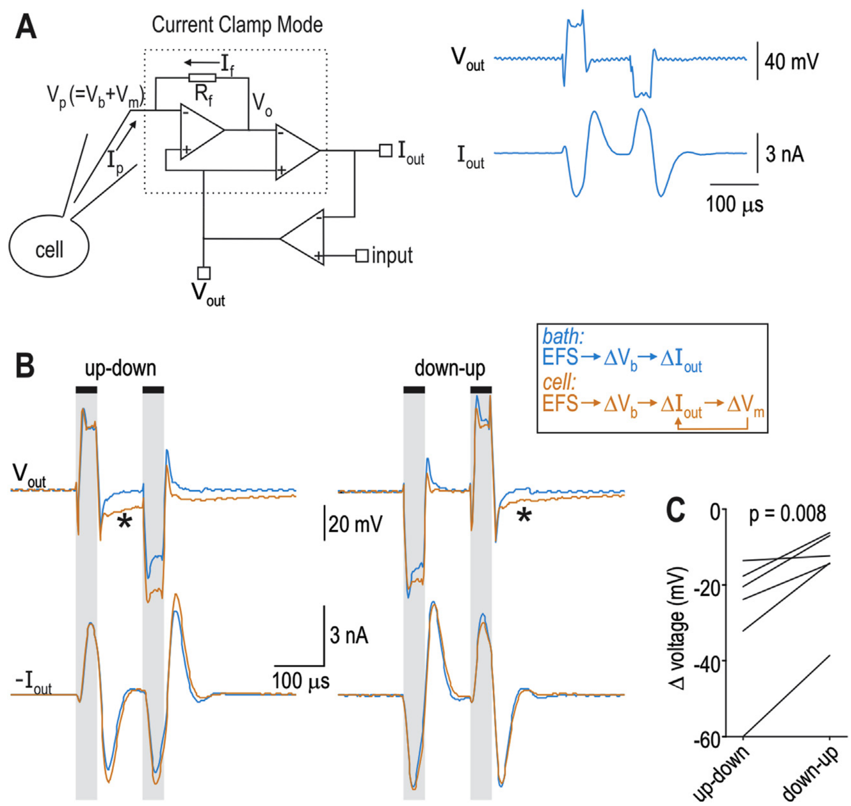

Fig. 8. EFS pulses induce an error current in PCAs that corrupts voltage measurement. (A) Schematic of PCA circuitry showing the *I-V* converter (inside dashed box) and the additional feedback loop required for current clamp. Based on the resistor feedback, pipette current (I_p) equals feedback current (I_f) and $V_p - V_0 = I_f R_f$. In voltage clamp, output current (I_{out}) represents the condition for pipette voltage (V_p) to equal command voltage (labeled here as output voltage, V_{out}). In current clamp, the difference between I_{out} and input is used to adjust V_p to maintain the desired current. Because V_p is the sum of bath voltage (V_b) and membrane voltage (V_m), changes in V_b driven by EFS pulses cause fluctuations in I_{out} (see traces on right), but in current clamp, fluctuations in I_{out} feedback onto V_m. Since the inverse of I_{out} is delivered back to the pipette [24], I_{out} traces in panel B are shown inverted to facilitate understanding of their impact of V_m. **(B)** Sample traces show deviations (*) between V_{out} recorded in the bath (blue) and in whole-cell (orange) because of feedback between I_{out} and V_m in the latter case. These deviations, which correspond to the unitary hyperpolarization, occurred selectively after upward voltage deflections and were larger when upward deflections preceded downward deflections (labeled up-down) than the opposite (down-up). **(C)** The differential unitary hyperpolarization observed in B predicted that kEFS associated with an up-down voltage deflection should yield greater cumulative hyperpolarization than that associated with a down-up voltage deflection. Testing in 6 neurons with 3 mA, 1 kHz EFS confirmed this ($t = 4.26$, $p = .008$, paired *t*-test).

nanoamperes of current delivered to the bath via the recording pipette (when the pipette is not patched on a neuron) will not affect bath voltage (V_b) – orders of magnitude stronger current (i.e. milliamperes) are delivered by the stimulator to the bath. On the other hand, a few nanoamperes delivered directly to the inside of a neuron (during whole cell recording) can dramatically alter membrane voltage (V_m). Changes in V_m are slowed by the membrane capacitance, which is notable since the 40 µs-long EFS pulses are very short relative to the membrane time constant, which is tens of milliseconds. Together these factors predict that voltage changes recorded in whole-cell mode (where pipette voltage $V_p = V_b + V_m$) will deviate slowly from the otherwise abrupt voltage changes recorded in the bath (where $V_p = V_b$) because, in the former case, I_{out} will affect V_m and vice versa, thus creating a feedback loop through which V_m is unrecoverably altered (see box in Fig. 8B).

Sample traces in Fig. 8B confirm this prediction. In all neurons tested ($n = 6$), voltage deviations indicative of altered V_m (see *s on Fig. 8B) appeared after an upward voltage deflection and were greater when the upward deflection preceded the downward deflection during biphasic EFS. This difference in unitary hyperpolarization predicts that cumulative hyperpolarization will differ depending on whether EFS evokes up-down or down-up voltage deflections, which depends on the anode-cathode designation on EFS leads and the neuron's position in the electric field. The prediction was confirmed (Fig. 8C, $t = 4.26$, $p = .008$, paired *t*-test). The basis for this asymmetry remains unresolved, but our data clearly demonstrate that EFS can cause PCAs to inject current that corrupts the measurement of cell voltage.

Discussion

Our results demonstrate that kEFS does not induce hyperpolarization that could account for the analgesic effects of high-rate SCS. Specifically, kEFS hyperpolarizes neurons only when they are recorded with a PCA, which is obviously not the conditions under which neurons normally operate. According to VSD imaging, unpatched neurons were not hyperpolarized during kEFS (Fig. 4), and nor was any kEFS-induced hyperpolarization observed in neurons recorded with a VFA (Fig. 5 and Fig. 6). Further testing uncovered an artifact linked to the current clamp mode of PCAs. The lack of kEFS-induced hyperpolarization – after excluding the artifact – is consistent with past studies (see Introduction) showing that kEFS is more likely to cause *de*polarization, and only when applied at frequencies and amplitudes higher than those used here or by Lee et al. [21]. Beyond disproving kEFS-based direct hyperpolarization as a novel mode of action for high-rate SCS, our results raise important concerns about artifacts that arise during extracellular EFS.

Magistretti and colleagues previously raised concerns about current clamp recordings conducted with PCAs. Their first report emphasized the mismeasurement of membrane voltage [23], but subsequent work clarified that voltage was actually perturbed during the measurement process [24]. This constitutes an *observer effect*: Whereas a simple mismeasurement caused by the quasi-static superposition of fluctuations in V_b on top of an otherwise accurate signal representing V_m could be corrected post-recording (e.g. by subtracting changes in V_b from V_p to isolate changes in V_m), the introduction of an artifactual current during the recording process introduces *un*correctable changes in the voltage since nonlinearities arise from the numerous voltage-gated ion channels in the cell membrane. The problem occurs because PCAs are optimized for low-noise voltage clamp recording. Current clamping is achieved in PCAs in a rather convoluted way, which, as we have shown, can be compromised by extracellular EFS. Specifically, when EFS-induced changes in I_{out} feed back onto V_m, voltage measurement is compromised in a subtle yet irreversible way (unlike the simple superposition of fluctuations in V_b on top of an otherwise accurate signal representing V_m).

The impact on V_m of ultra-short fluctuations in I_{out} is small (Fig. 8) and has, therefore, likely gone unrecognized in past experiments that utilized EFS in conjunction with PCA-based current clamp recordings. However, if I_{out} fluctuations recur before the unitary hyperpolarization has dissipated, the effects summate and cause substantial cumulative hyperpolarization (Fig. 7). When EFS is applied at kilohertz frequencies, the cumulative hyperpolarization became obvious, and suspicious. We focussed on summation occurring at 1 kHz EFS but, at higher rates, even faster voltage fluctuations may summate, which likely explains the strong but highly variable voltage responses observed during 5 and 10 kHz EFS. But even when single pulses are used, PCA-based voltage measurements during or shortly after the pulse are likely to be compromised. Separate from measuring voltage, other measurements like rheobase and chronaxie would also be affected by the introduction of artifactual current from the PCA during extracellular stimulation. It is difficult to judge the degree to which past studies may have been affected because there are numerous factors (e.g. pipette shielding) that impact the degree to which this artifact is manifest, but preventive steps should be taken to avoid this artifact in future studies.

Conclusions

The biophysical mechanism underlying the analgesic effect of high-rate SCS remains unclear. According to simulations, nerve block during high-rate SCS required exceedingly strong stimulus intensities [22] and transient neuronal activation typically precedes the nerve block induced by kEFS [5], yet there is no clinical evidence for transient activation during high-rate SCS [20]. Furthermore, results from the Evaluation of Spinal Cord Stimulation Pulse Rate On Clinical Outcomes (PROCO) clinical trial suggest that SCS at rates between 1 and 10 kHz provides equivalent pain relief [32], yet the bottom end of that range falls well outside the frequency range associated with nerve block. Beyond the presence or absence of paresthesia, other differences between low- and high-rate SCS such as optimal lead placement [33] hint at the differences in mechanism but do not point to obvious candidate mechanisms [20]. The rise and fall of direct inhibition via kEFS-induced hyperpolarization as a completely novel mode of action should serve as a cautionary tale about artifacts caused by electric fields. Vigilance against such artifacts is crucial for neuromodulation research and other fields utilizing electrical stimulation. At the very least, PCAs should be avoided (or used with caution) when conducting current clamp recordings when EFS is involved, especially if the goal is to measure the effects of that stimulation. More generally, independent approaches (e.g. VSD imaging) and technology (e.g. amplifiers with different designs) should be used to cross-validate results.

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Authorship statement

LSL and SR conducted experiments. ML and TZ conducted simulations. LSL, ML, TZ, SR and SAP analyzed data. All authors contributed to the study design and manuscript preparation.

Conflicts of interest

SAP is on the Scientific Advisory Board and TZ and RE are paid employees of Boston Scientific.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.brs.2017.12.004.

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